

# Dynamic Portfolio Optimization in the Saudi Stock Market: An RL Approach

## Leveraging LLM Sentiment and GRU Forecasting Signals

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### Introduction

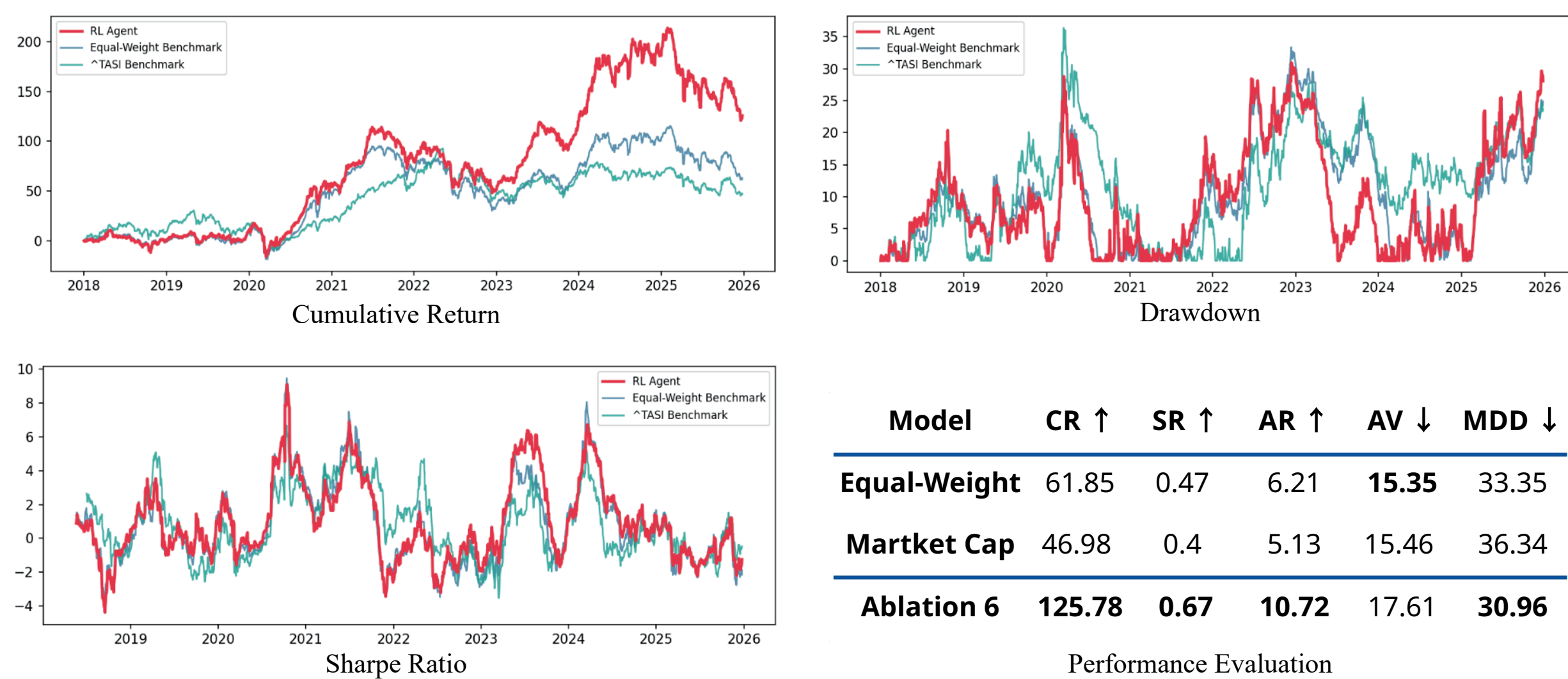
The increasing complexity of financial markets, fueled by rapid data growth and global economic shifts, has intensified the need for advanced portfolio optimization that balances risk and return. While traditional trading relies on numerical data, qualitative news sentiment is now recognized as a vital driver of market dynamics.

Despite the Saudi Stock Exchange (Tadawul) being the largest in the Middle East, there is a significant research gap in integrating modern AI-driven strategies within this specific market. This study addresses this by developing a framework that leverages Large Language Models (LLMs) for sentiment extraction, Gated Recurrent Units (GRU) for return forecasting, and Reinforcement Learning (RL) for adaptive decision-making.

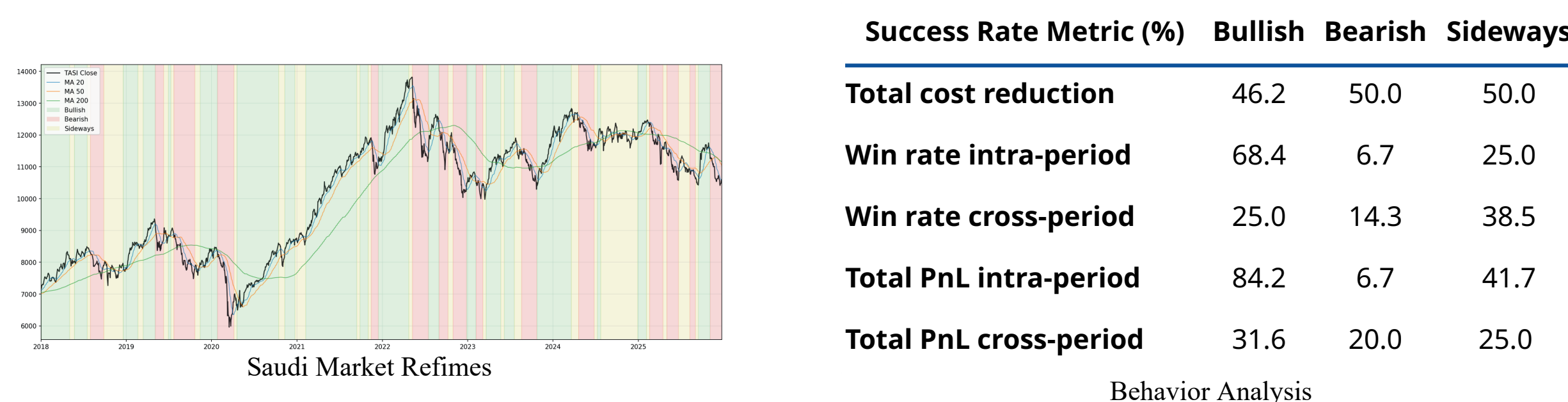
#### Main Objectives:

- Develop an RL Framework:** Build a framework for dynamic portfolio optimization and adaptive trading for the Saudi stock market.
- Signal Integration:** Combine GRU-based forecasting and LLM-based sentiment analysis to enhance market awareness and portfolio allocation.
- Performance Evaluation:** Analyze the impact of these integrated signals on trading performance against baseline portfolio strategies.
- Behavioral Analysis:** Analyze the agent's behavior across different market regimes to assess its adaptability and decision-making patterns under varying market conditions.
- Ablation Analysis:** Evaluate each signal's contribution and test whether LLM sentiment and GRU predictions are more effective as observations or reward components.

### Experiments and Results

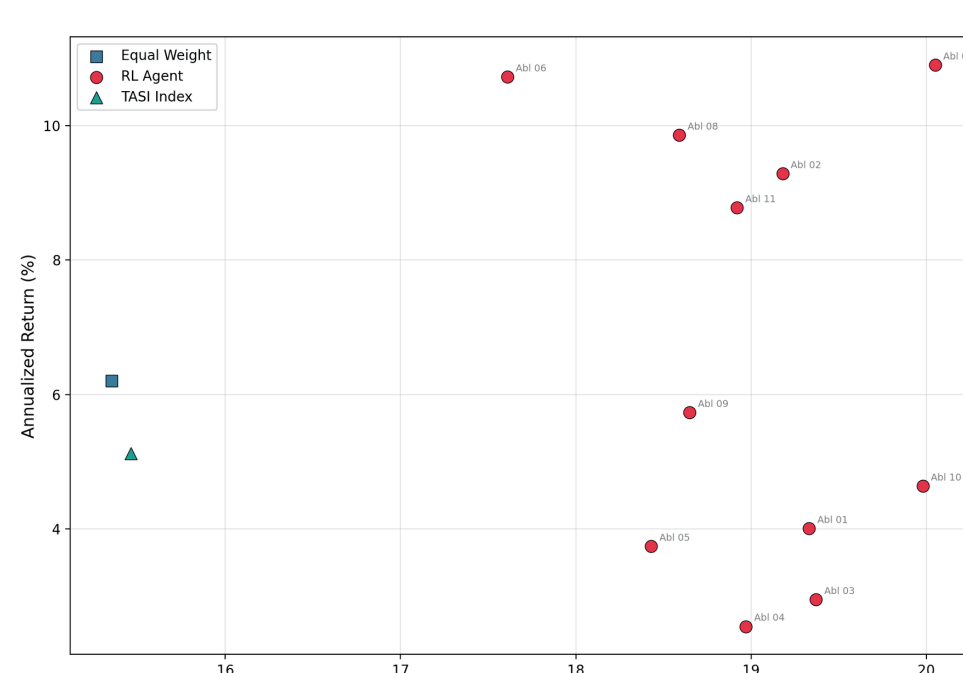


The RL agent outperforms both passive benchmarks on every return-based metric while delivering the lowest maximum drawdown of the three strategies. Over the 2018–2026 horizon, it more than doubles the equal-weight portfolio's cumulative return and achieves a Sharpe ratio of 0.67 against 0.47 for the benchmark.



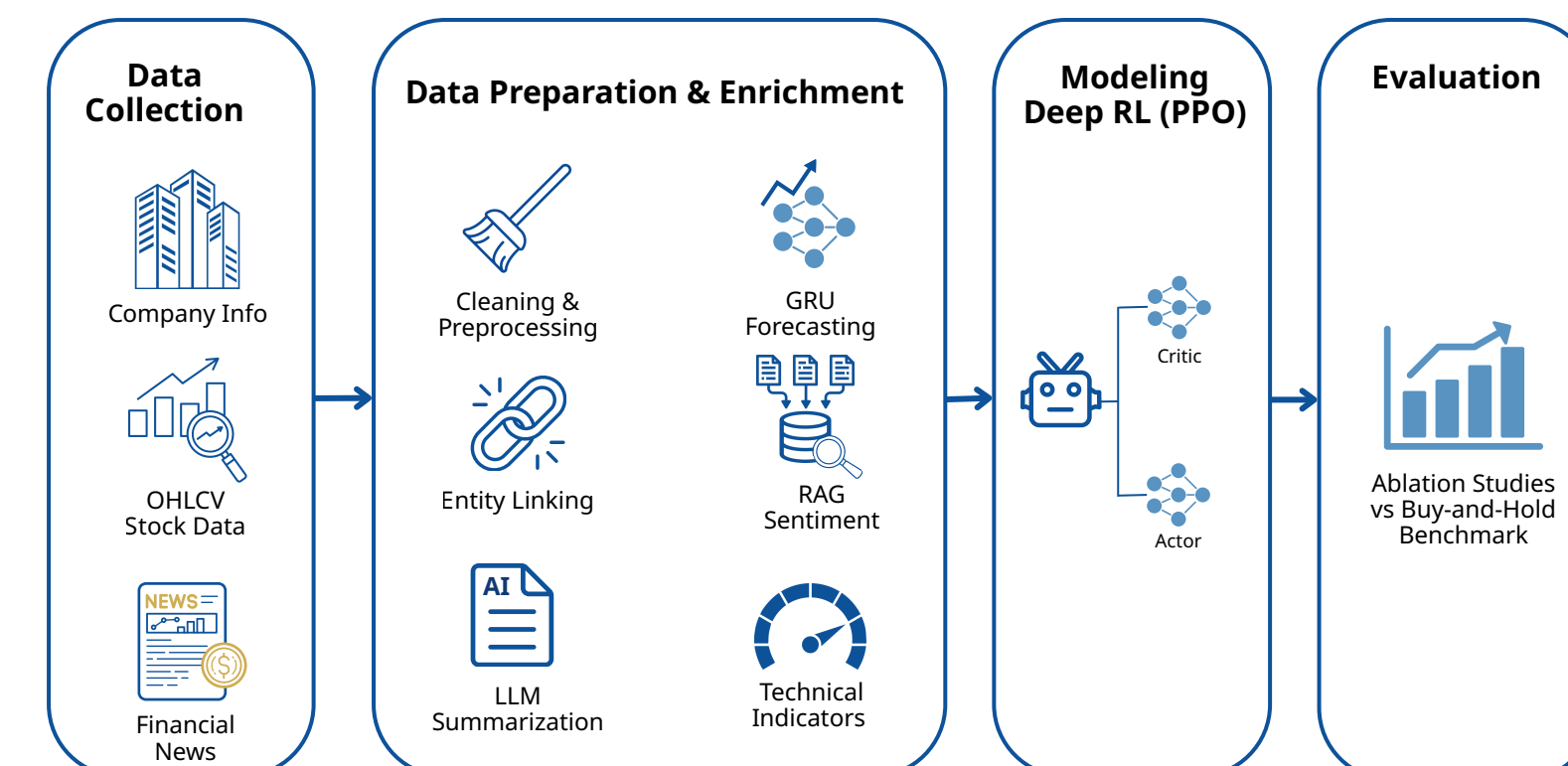
A period is a continuous span of a market regime. Intra-period metrics evaluate performance within a single regime period, whereas cross-period metrics evaluate performance across regime transitions. As most periods are short, intra-period outcomes are more likely to remain within the agent's observation window, while cross-period outcomes more often extend beyond it. Performance is regime-dependent, with bullish and sideways regimes showing consistent profitability patterns, whereas bearish regimes show deterioration across nearly all metrics. However, cost reduction remains consistently around 50% across all regimes.

Ablation studies were conducted to evaluate the impact of return prediction and sentiment signals on model performance, and to determine whether they are more effective when used as observations, reward components, both, or neither.

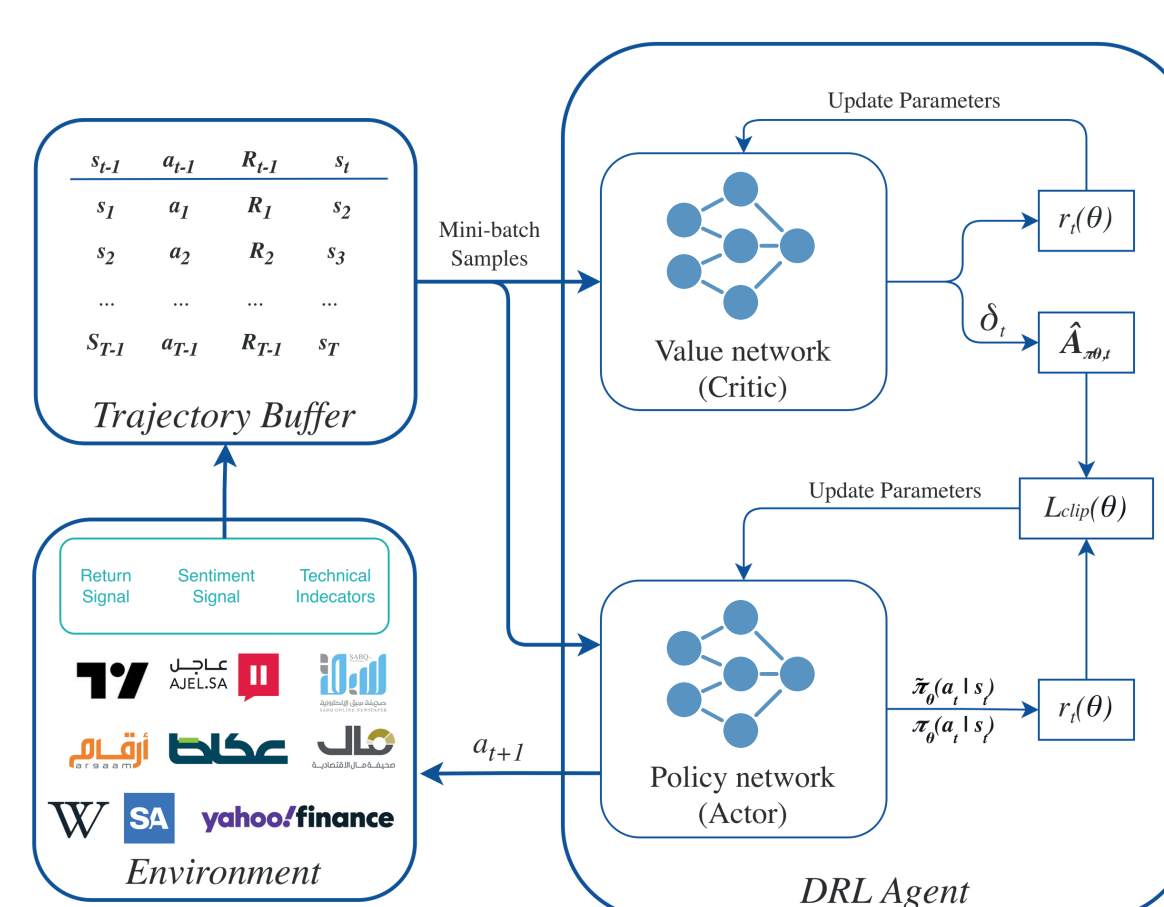


### Methodology

A DRL agent makes Buy, Sell, and Hold decisions, guided by a multi-objective reward balancing profitability and risk with a 3-day trade cooldown constraint.



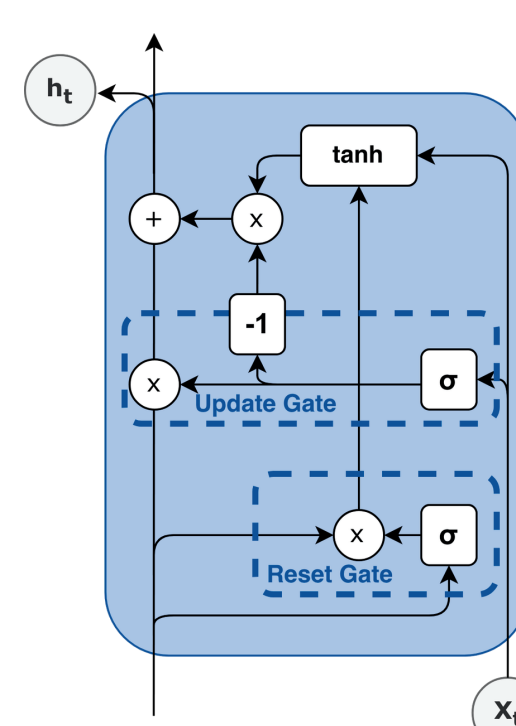
- Data collection:** Asynchronously fetched 320,106 news articles and 755,876 OHLCV records covering the Saudi market.
- Entity linking & summarization:** Company names are matched across metadata fields, and articles are condensed into 33,476 summaries.
- Technical indicators:** RSI, volatility, and z-scores capture momentum, and statistical price deviations are calculated to help the model recognize complex market patterns.
- Return forecasting:** A GRU model predicts returns by learning long-term price dependencies through its memory mechanism.
- Sentiment Analysis:** Retrieval-Augmented Generation (RAG) provides the LLM with 10 historical contexts per summary to generate a sentiment score that is either positive, negative, or neutral.



Proximal Policy Optimization (PPO) is an on-policy reinforcement learning algorithm designed to provide stable policy updates, preventing an agent from falling “off the cliff”. PPO balances exploration and exploitation through stochastic policy sampling, gradually shifting toward exploitation during training. To ensure prevention of violated actions, **masking** is applied to force the

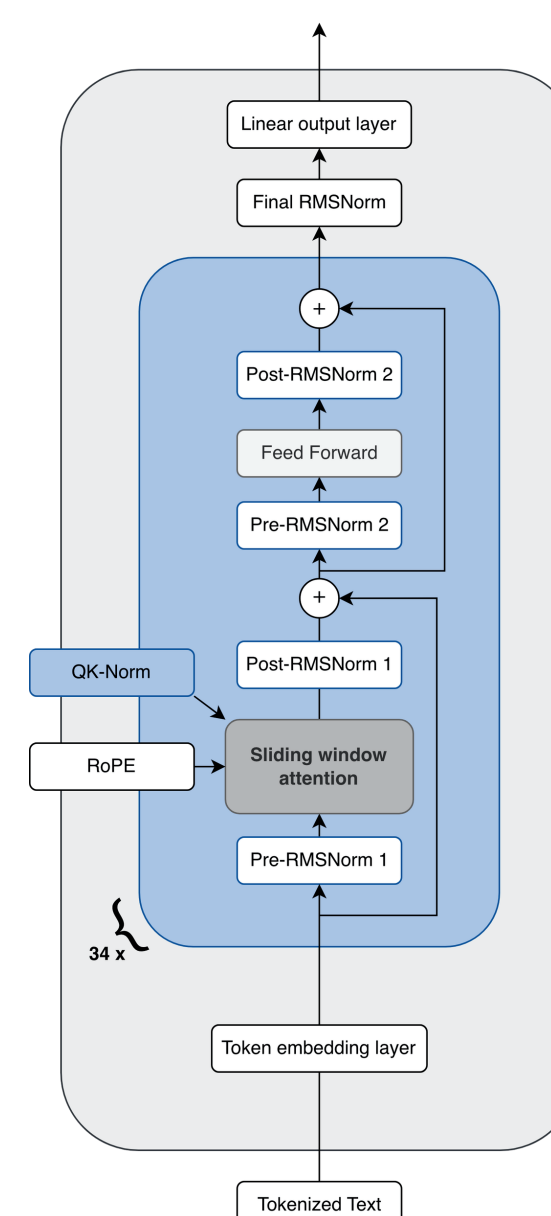
model to ignore invalid choices by setting their logits to  $-\infty$ , which results in a zero probability. PPO updates the policy based on a designed trading reward.

$$\text{Reward} = 0.5 \text{ Daily Return} + 0.5 \text{ EWMA VaR} - 0.02 \text{ Drawdown increase} - 0.03 \text{ Turnover} + 0.2 \text{ Sentiment} + 0.2 \text{ Return Forecast}$$



Gemma 3n is a resource-efficient variant of the Gemma 3 LLM that combines multimodal and multilingual capabilities. It achieves high efficiency through Per-Layer Embedding (PLE) and the Matryoshka Transformer (MatFormer) architecture, allowing it to dynamically partition its shared transformer into smaller, independent slices for flexible inference.

The GRU architecture is a simplified variant of the Long Short-Term Memory (LSTM) network that uses an update gate, which controls how much past information to retain, and a reset gate, which determines how much to forget, while maintaining a single hidden state to efficiently capture long-term dependencies with fewer parameters and faster training.



Commit to the instructions:

- Provide a concise summary.
  - Don't extend.
  - The summary length must be less than the whole content.
  - The summary must be in the context of the described company, neglecting any irrelevant details.
    - Description of {company\_name}: {company\_description}
  - Description Ends Here.
  - If the news doesn't match the description, then you must answer with "0" without summary.
- Articles Start Here:  
{news}

Answer the question based on the following context:

{context}

Question: What is the Sentiment of the next Article (negative/neutral/positive), don't extend:  
{summary}

Answer:

### Conclusion

The proposed RL framework successfully integrates LLM-based sentiment analysis and GRU return forecasting for dynamic portfolio optimization in the Saudi stock market. Experimental results show that the PPO agent outperforms passive benchmark strategies across return and risk metrics while maintaining lower drawdown. One **limitation** is that the model struggles in bearish markets, where profitability and win rates drop despite success in other markets. **Future work** could address this by replacing the single-policy PPO with a regime-aware hierarchical or multi-agent framework. This would use a gating mechanism to dynamically switch between specialized policies based on detected market conditions.